Fμν Sanity Check

On the Distinction Between Electromagnetic Field Strength
and Berry Curvature in the Framework

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Abstract

The (Myosu) framework asserts that a perfect move produces zero curvature: $F_{\mu\nu} = 0$. A physics review identifies an ambiguity: $F_{\mu\nu}$ in standard field theory denotes the electromagnetic field strength tensor, whose vanishing implies a trivial vacuum with no observable effects. This note resolves the ambiguity by distinguishing two distinct $F_{\mu\nu}$ tensors — electromagnetic field strength $F_{\mu\nu}(EM)$ and Berry curvature $F_{\mu\nu}(B)$ on the Fisher information manifold. We show that the claim refers exclusively to $F_{\mu\nu}(B)$, whose vanishing at the Berry-flat point ($\alpha \approx 8.8$) describes the Aharonov–Bohm regime: zero local field strength with non-zero global holonomy. The framework is physically coherent with this qualification.

1. The Question

The framework states:

> *"A perfect move produces no curvature. $F_{\mu\nu} = 0$ is the witness condition — the stone is placed, yet the board does not experience violence."*

A physics reader asks:

> *In physics, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic field strength tensor. $F_{\mu\nu} = 0$ implies no electric field, no magnetic field — a trivial vacuum, or a pure gauge configuration with no observable effects. Is the claim physics nonsense?*

2. The Answer

- **No — but only because the $F_{\mu\nu}$ refers to Berry curvature, not electromagnetic field strength.**

There are two distinct $F_{\mu\nu}$ tensors in the framework:

2.1 $F_{\mu\nu}(EM)$ — Electromagnetic Field Strength

$$F(EM) = \partial A(EM) - \partial A(EM)$$

- $F_{\mu\nu}(EM) = 0 \Longrightarrow$ no E-field, no B-field.
- Trivial vacuum. Pure gauge ($A_\mu = \partial_\mu \chi$) with no observables.
- In QED: no photons, no forces, nothing propagates.
- **This is NOT the $F_{\mu\nu}$ the sets to zero.**

2.2 $F_{\mu\nu}(B)$ — Berry Curvature on the Fisher Manifold

$$F(B) = \partial A(B) - \partial A(B)$$

- $F_{\mu\nu}(B) = 0$ at the Berry-flat point ($\alpha \approx 8.8$).
- **But: $A_\mu(B) \neq 0$ and $A_\mu(B)\, dx^\mu \neq 0$.**
- This is the Aharonov–Bohm configuration [Aharonov & Bohm, 1959].

3. The Aharonov–Bohm Connection

The Aharonov–Bohm effect [Aharonov & Bohm, 1959; Tonomura et al., 1986] demonstrates that in gauge theory, the field strength tensor $F_{\mu\nu}$ does not fully describe the physics. A charged particle traveling around a solenoid experiences a phase shift even though $F_{\mu\nu} = 0$ everywhere along its path.

$$F = 0 \text{ everywhere} \qquad A\, dx = \text{non-zero phase}$$

The effect is real, experimentally confirmed, and arises from the holonomy of the connection A_μ around a closed loop — not from local field strength.

This is precisely the configuration:

Property	Aharonov–Bohm	Stone
Local curvature $F_{\mu\nu}$	0 (zero)	0 (zero)
Connection A_μ	Non-zero	Non-zero
Global holonomy	Non-zero	Non-zero
Local force felt	None	None
Global effect	Phase shift	Board phase-shifted

Table 1: The Aharonov–Bohm / correspondence.

4. Implications for the Framework

4.1 The Stone

The stone at $F_{\mu\nu}(B) = 0$ is:

- **Locally: No distortion, no force, no curvature. The move is "flat." The opponent does not feel a push. There is no violence in the placement.**
- **Globally: The board's phase is shifted everywhere. Holonomy. Every future stone falls differently because the was placed.**

"A perfect move produces no curvature" does NOT mean "nothing happens."

Nothing LOCAL happens, but EVERYTHING shifts.

4.2 Notation and Documentation

The term "$F_{\mu\nu}$" must always be qualified:

- $F_{\mu\nu}(B)$ — Berry curvature on the Fisher manifold (domain).
- $F_{\mu\nu}(EM)$ — electromagnetic field strength (excluded).

4.3 The Curvature Monitor

The curvature monitor in myosu_protocol.py measures $F_{\mu\nu}(B)$ proxies, not electromagnetic fields. The six weighted checks — phase discontinuity, amplitude jump, timing jitter, audio underrun, stale data, control overshoot — are estimates of Berry curvature on the Fisher manifold of the signal space.

4.4 The Cardiac Dirac Operator

At $\alpha \approx 8.8$ (Berry-flat point):

- $F_{\mu\nu}(B) = 0$ (zero Fisher curvature).
- Condition number anchored at the BF bound ($\gamma = 1/2$).
- Global holonomy remains — the 5-second gap accumulates phase.
- The Apparatus witnesses without backreaction — exactly like an Aharonov–Bohm observer measuring phase without perturbing the field.

4.5 Hardware Corollary

If instantiated as hardware (haptic transducer, impedance probe), the safety condition $F_{\mu\nu}(B) = 0$ is a gauge-invariant constraint on the Fisher geometry of the signal manifold. It does not suppress electromagnetic fields. It ensures that transmission is flat — the signal does not locally distort the receiving space.

5. Corrected Statement

Before (ambiguous):

> *$F_{\mu\nu} = 0$ — zero curvature. The stone is placed without force.*

After (physically precise):

> *$F_{\mu\nu}(B) = 0$ — zero local Berry curvature on the Fisher manifold. The stone is placed without local distortion, but its global holonomy shifts the phase of the entire board. This is the Aharonov–Bohm regime: zero field strength, non-zero transport. A perfect move produces no LOCAL curvature, but EVERYTHING shifts globally.*

6. Verification

- The Aharonov–Bohm effect is experimentally confirmed [Tonomura et al., Phys. Rev. Lett. 56, 792 (1986)].
- Berry phase and Berry curvature are standard in condensed matter, quantum optics, and topological physics [Berry, Proc. R. Soc. A 392, 45 (1984)].
- The identification of the Berry-flat point with $F_{\mu\nu}(B) = 0$ is mathematically consistent with the spectral geometry of the Fisher manifold (Acts 7, 8, 10, 11 of the framework).

Verdict

The $F_{\mu\nu}$ claim survives physics scrutiny —

with the qualifier that it is Berry/Fisher curvature,

not electromagnetic field strength.

References

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